

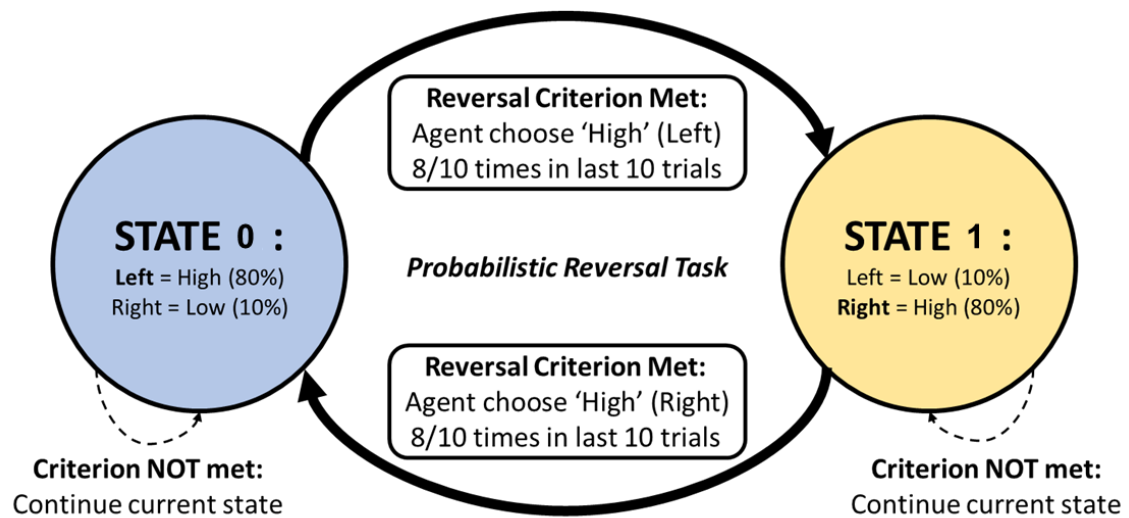
Computational Models

$z \in \{0,1\}$, z is a state. $z = 0$, when left is high reward prob. and $z = 1$, when right is high reward prob.

high reward prob. = 0.8

low reward prob. = 0.1

Probabilistic Reversal Learning Task:



Bayesian Inference model:

“Probability Theory”

Fitting parameters: $[P(\text{rev}), \gamma, \beta, \beta_0]$ (reversal probability, forgetting factor, inverse temperature term, bias)

- Belief Tracking: tracks the probability of two states (right high reward prob [RH], left high reward prob [LH])
- Inference: uses Bayes' Theorem to update the belief based on the outcome (rewarded or not) each trial
- Reversal Expectation: parameter $p(\text{reversal})$ accounts for the possibility that the reward contingencies flipped between trials
- Q-estimation: chosen action's value is calculated by weighting the fixed reward probabilities by the belief state which unchosen action is decayed using parameter γ

$b_t = P(r | z=1) b_t / (P(r | z=1) b_t + P(r | z=0) (1 - b_t))$ # Bayes' Theorem, b_t = belief state of right reward prob. high

$b_t = (1 - P(\text{rev})) b_t + P(\text{rev})(1 - b_t)$ #Reversal probability

$Q_{t,a} = 0.8 b_t + 0.1 (1 - b_t)$, if $a = \text{right}$ #chosen quality if action is right

$Q_{t,a} = 0.1 b_t + 0.8 (1 - b_t)$, if $a = \text{left}$ #chosen quality if action is left

$Q_{t,a'} = \gamma Q_{t-1,a'}$ #unchosen quality

Reinforcement Learning model:

“Trial & Error”

Fitting parameters: $[a, \gamma, \beta, \beta_0]$ (learning rate, decay, inverse temperature term, bias)

- Update Rule: uses Delta rule ($Q_t = Q_{t-1} + a(\text{RPE})$)
- Decaying Memory: γ^{t-k} term in RPE, which suggests that the value of an action decays if it hasn't been chosen for a long time (t = current trial number, k = elapsed time (trials numbers) since that specific action was last performed)

$\text{RPE}_{t,a} = r_{t,a} - \gamma^{t-k} Q_{k,a}$ #Reward Prediction Error

$Q_{t,a} = Q_{t-a,a} + a(\text{RPE})$ #Delta rule

RL model with Uncertainty Bonus

“Strategic Exploration”

Fitting parameters: $[a, \gamma, c, \beta, \beta_0]$ (learning rate, confident constant, decay, inverse temperature term, bias)

- Uncertainty Bonus (UB): in addition to the RL, the model calculates a bonus based on how rarely an action has been chosen ($N(a)$) relative to the total trials (T) and $N(a)$ and T reset back to 1 when difference of two state probabilities are less than 0.2
- Confidence Constant (c): this parameter pushes the agent to choose actions that is uncertain about, encouraging information gain

$\text{RPE}_{t,a} = r_{t,a} - \gamma^{t-k} Q_{k,a}$

$Q_{t,a} = Q_{t-a,a} + a(\text{RPE})$

$\text{UB} = \sqrt{\frac{\ln(T)}{N(a)}}$ #Uncertainty Bonus

Belief RL model:

“Hybrid”

Fitting parameters: $[a, q, \beta, \beta_0]$ (learning rate, switch probability, inverse temperature term, bias)

- Weighting Beliefs: instead of estimating values for actions (left or right), model calculates weights (w_1, w_2) for different hypotheses (e.g., "State RH is active" or "State LH is active")
- Q-estimation: quality is the sum of these weights, each multiplied by the current probability that that hypothesis is true
- Learning: weights (w_1, w_2) are updated by the RPE, but the update is scaled by the belief

$Q_{t,a} = w_1 b_t(a) + w_2 b_t(a')$ # b_t = belief states of left ($z=0$) and right ($z=1$)

$RPE_{t,a} = r_{t,a} - Q_{t,a}$

$w_1 = w_1 + a RPE_{t,a} b_t(a)$ # weights update

$w_2 = w_2 + a RPE_{t,a} b_t(a')$

$L(z) = P_1$, if $a = z$ and $r_t = 1$ # $L(z)$ = likelihoods of two states ($z=0, z=1$)

$L(z) = 1 - P_1$, if $a = z$ and $r_t = 0$

$L(z) = P_2$, if $a \neq z$ and $r_t = 1$

$L(z) = 1 - P_2$, if $a \neq z$ and $r_t = 0$

$b_t(z) = \frac{L(z) b_t(z)}{\sum_{j \in \{0,1\}} L(j) b_t(j)}$ # Bayes' Theorem

$b_{t+1}(z) = (1-q) b_t(z) + q b_t(1-z)$ # Switch Probability

Pearce-Hall model:

“Adaptative Learning”

Fitting parameters: [a_{positive} , a_{negative} , γ , ψ , β , β_0] (learning rates (+&-), forgetting term, surprising term, inverse temperature term, bias)

- Adaptive learning (a_v): this dynamic variable changes based on how “surprised” the agent is
- Surprise-Driven Learning: if agent experience large RPEs, a_v increases, making the agent learn faster to adapt to the potential change and as environment becomes more predictable, a_v decreases
- Asymmetric learning: separate learning rate for positive and negative RPE (a_{positive} , a_{negative})

$$\text{RPE}_{t,a} = r_{t,a} - Q_{t,a}$$

$$a = a_{\text{positive}}, \text{ if } \text{RPE}_{t,a} \geq 0 \text{ \#asymmetric learning rates}$$

$$a = a_{\text{negative}}, \text{ if } \text{RPE}_{t,a} < 0$$

$$Q_{t,a} = Q_{t-1,a} + (a * a_v * \text{RPE}_{t,a})$$

$$Q_{t,a} = Q_0 + \gamma * (Q_{t-1,a} - Q_0) \text{ \# } Q_0 = 0.5$$

$$a_v = a_v + \psi(|\text{RPE}_{t,a}| - a_v) \text{ \# } a_v \text{ update}$$

Action Probability (sigmoid function):

To estimate action probability from quality, sigmoid function is used.

$$\Delta Q = Q_r - Q_l$$

$$P_{(r)} = 1/(1+\exp(-(\beta * \Delta Q + \beta_0))) \text{ \#right choice prob.}$$

$$P_{(l)} = 1 - P_{(r)} \text{ \#left choice prob.}$$

To incorporate Uncertainty Bonus in **RL model with Uncertainty Bonus**:

$$\Delta Q = Q_r - Q_l$$

$$P_{(r)} = 1/(1+\exp(-(\beta * \Delta Q + c * \Delta U B + \beta_0))) \text{ \#right choice prob.}$$

$$P_{(l)} = 1 - P_{(r)} \text{ \#left choice prob.}$$

Logistic Regression:

To see how much a subject's current choice is driven by "Success" (Reward) vs. "Failure" (No Reward) as well as to measure memory decay by showing whether decisions rely on immediate trials or a longer history of events.

$$\text{logit}(P_{(i)}) = \sum_{i=1}^{10} \beta_{\text{Rew}} * \text{RewC}(t-i) + \sum_{i=1}^{10} \beta_{\text{Unr}} * \text{UnrC}(t-i) + \beta_0$$

RewC = rewarded choices

UnrC = unrewarded choices